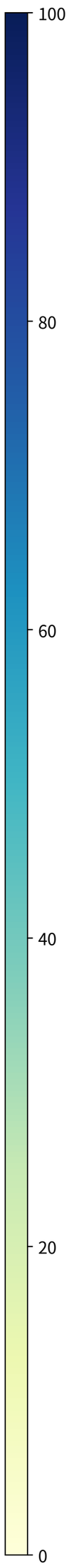
Boku-nano 18モデル｜middle4追加後のカテゴリ別pass@1 / pass@5

pass@1：greedy 1候補

	1操作への転移 (n=96)	未学習2操作 (n=100)	3→2操作への転移 (n=88)	同一操作×2 (n=96)	未学習3操作 (n=100)	同一操作×3 (n=96)	一般4操作 (n=100)	ABAB型4操作 (n=100)	同一操作×4 (n=96)
15M・1 epoch	88/96 91.7%	98/100 98.0%	88/88 100.0%	94/96 97.9%	96/100 96.0%	94/96 97.9%	29/100 29.0%	70/100 70.0%	71/96 74.0%
15M・3 epoch	96/96 100.0%	98/100 98.0%	88/88 100.0%	95/96 99.0%	100/100 100.0%	91/96 94.8%	2/100 2.0%	56/100 56.0%	61/96 63.5%
15M・10 epoch	95/96 99.0%	100/100 100.0%	87/88 98.9%	79/96 82.3%	99/100 99.0%	70/96 72.9%	0/100 0.0%	48/100 48.0%	34/96 35.4%
5M・1 epoch	81/96 84.4%	94/100 94.0%	85/88 96.6%	89/96 92.7%	95/100 95.0%	91/96 94.8%	7/100 7.0%	62/100 62.0%	72/96 75.0%
1M・3 epoch	82/96 85.4%	99/100 99.0%	88/88 100.0%	96/96 100.0%	100/100 100.0%	95/96 99.0%	43/100 43.0%	78/100 78.0%	70/96 72.9%
15M・1 epoch［短いBPE］	96/96 100.0%	100/100 100.0%	87/88 98.9%	94/96 97.9%	99/100 99.0%	91/96 94.8%	22/100 22.0%	74/100 74.0%	58/96 60.4%
5M・1 epoch［短いBPE］	93/96 96.9%	100/100 100.0%	86/88 97.7%	96/96 100.0%	96/100 96.0%	95/96 99.0%	14/100 14.0%	73/100 73.0%	72/96 75.0%
1M・3 epoch［短いBPE］	94/96 97.9%	100/100 100.0%	88/88 100.0%	96/96 100.0%	100/100 100.0%	95/96 99.0%	15/100 15.0%	73/100 73.0%	63/96 65.6%
6層・幅384 / 12.19M・1 epoch	86/96 89.6%	100/100 100.0%	88/88 100.0%	93/96 96.9%	99/100 99.0%	96/96 100.0%	26/100 26.0%	75/100 75.0%	80/96 83.3%
8層・幅256 / 9.44M・1 epoch	87/96 90.6%	97/100 97.0%	88/88 100.0%	94/96 97.9%	99/100 99.0%	95/96 99.0%	33/100 33.0%	77/100 77.0%	79/96 82.3%
4層・幅384 / 8.65M・1 epoch	82/96 85.4%	100/100 100.0%	88/88 100.0%	94/96 97.9%	100/100 100.0%	92/96 95.8%	24/100 24.0%	67/100 67.0%	61/96 63.5%
2層・幅384 / 5.11M・1 epoch	80/96 83.3%	96/100 96.0%	84/88 95.5%	96/96 100.0%	96/100 96.0%	95/96 99.0%	5/100 5.0%	54/100 54.0%	61/96 63.5%
中間層×4・残差1 / 1M・3 epoch	75/96 78.1%	96/100 96.0%	87/88 98.9%	93/96 96.9%	98/100 98.0%	95/96 99.0%	53/100 53.0%	77/100 77.0%	74/96 77.1%
中間層×4・残差0.25 / 1M・3 epoch	83/96 86.5%	98/100 98.0%	86/88 97.7%	95/96 99.0%	99/100 99.0%	94/96 97.9%	37/100 37.0%	70/100 70.0%	68/96 70.8%
LT1 0.426M・3 epoch［設定復元］	36/96 37.5%	33/100 33.0%	31/88 35.2%	52/96 54.2%	18/100 18.0%	33/96 34.4%	0/100 0.0%	19/100 19.0%	21/96 21.9%
F128 0.377M・3 epoch［設定復元］	24/96 25.0%	21/100 21.0%	20/88 22.7%	33/96 34.4%	7/100 7.0%	20/96 20.8%	0/100 0.0%	12/100 12.0%	10/96 10.4%
W96 0.289M・3 epoch［設定復元］	4/96 4.2%	0/100 0.0%	0/88 0.0%	4/96 4.2%	0/100 0.0%	4/96 4.2%	0/100 0.0%	0/100 0.0%	2/96 2.1%
W64 0.172M・3 epoch［設定復元］	0/96 0.0%	0/100 0.0%	0/88 0.0%	1/96 1.0%	0/100 0.0%	1/96 1.0%	0/100 0.0%	0/100 0.0%	0/96 0.0%

pass@5：greedy 1＋sampling 4候補

	1操作への転移 (n=96)	未学習2操作 (n=100)	3→2操作への転移 (n=88)	同一操作×2 (n=96)	未学習3操作 (n=100)	同一操作×3 (n=96)	一般4操作 (n=100)	ABAB型4操作 (n=100)	同一操作×4 (n=96)
15M・1 epoch	95/96 99.0%	100/100 100.0%	88/88 100.0%	96/96 100.0%	99/100 99.0%	96/96 100.0%	75/100 75.0%	93/100 93.0%	88/96 91.7%
15M・3 epoch	96/96 100.0%	100/100 100.0%	88/88 100.0%	96/96 100.0%	100/100 100.0%	94/96 97.9%	5/100 5.0%	65/100 65.0%	65/96 67.7%
15M・10 epoch	96/96 100.0%	100/100 100.0%	88/88 100.0%	91/96 94.8%	100/100 100.0%	85/96 88.5%	0/100 0.0%	66/100 66.0%	50/96 52.1%
5M・1 epoch	92/96 95.8%	98/100 98.0%	88/88 100.0%	96/96 100.0%	100/100 100.0%	95/96 99.0%	32/100 32.0%	78/100 78.0%	79/96 82.3%
1M・3 epoch	95/96 99.0%	100/100 100.0%	88/88 100.0%	96/96 100.0%	100/100 100.0%	96/96 100.0%	80/100 80.0%	93/100 93.0%	87/96 90.6%
15M・1 epoch［短いBPE］	96/96 100.0%	100/100 100.0%	88/88 100.0%	96/96 100.0%	100/100 100.0%	95/96 99.0%	79/100 79.0%	93/100 93.0%	84/96 87.5%
5M・1 epoch［短いBPE］	95/96 99.0%	100/100 100.0%	88/88 100.0%	96/96 100.0%	98/100 98.0%	96/96 100.0%	76/100 76.0%	93/100 93.0%	84/96 87.5%
1M・3 epoch［短いBPE］	96/96 100.0%	100/100 100.0%	88/88 100.0%	96/96 100.0%	100/100 100.0%	95/96 99.0%	55/100 55.0%	87/100 87.0%	76/96 79.2%
6層・幅384 / 12.19M・1 epoch	96/96 100.0%	100/100 100.0%	88/88 100.0%	96/96 100.0%	100/100 100.0%	96/96 100.0%	54/100 54.0%	88/100 88.0%	87/96 90.6%
8層・幅256 / 9.44M・1 epoch	96/96 100.0%	100/100 100.0%	88/88 100.0%	96/96 100.0%	100/100 100.0%	96/96 100.0%	66/100 66.0%	95/100 95.0%	87/96 90.6%
4層・幅384 / 8.65M・1 epoch	94/96 97.9%	100/100 100.0%	88/88 100.0%	96/96 100.0%	100/100 100.0%	96/96 100.0%	59/100 59.0%	89/100 89.0%	82/96 85.4%
2層・幅384 / 5.11M・1 epoch	94/96 97.9%	100/100 100.0%	88/88 100.0%	96/96 100.0%	100/100 100.0%	96/96 100.0%	28/100 28.0%	74/100 74.0%	75/96 78.1%
中間層×4・残差1 / 1M・3 epoch	94/96 97.9%	100/100 100.0%	88/88 100.0%	96/96 100.0%	100/100 100.0%	96/96 100.0%	88/100 88.0%	93/100 93.0%	91/96 94.8%
中間層×4・残差0.25 / 1M・3 epoch	96/96 100.0%	100/100 100.0%	88/88 100.0%	96/96 100.0%	100/100 100.0%	96/96 100.0%	77/100 77.0%	92/100 92.0%	86/96 89.6%
LT1 0.426M・3 epoch［設定復元］	62/96 64.6%	49/100 49.0%	47/88 53.4%	72/96 75.0%	27/100 27.0%	51/96 53.1%	0/100 0.0%	33/100 33.0%	36/96 37.5%
F128 0.377M・3 epoch［設定復元］	57/96 59.4%	38/100 38.0%	28/88 31.8%	52/96 54.2%	14/100 14.0%	33/96 34.4%	0/100 0.0%	19/100 19.0%	25/96 26.0%
W96 0.289M・3 epoch［設定復元］	11/96 11.5%	2/100 2.0%	5/88 5.7%	14/96 14.6%	0/100 0.0%	9/96 9.4%	0/100 0.0%	3/100 3.0%	8/96 8.3%
W64 0.172M・3 epoch［設定復元］	0/96 0.0%	0/100 0.0%	0/88 0.0%	4/96 4.2%	0/100 0.0%	2/96 2.1%	0/100 0.0%	1/100 1.0%	1/96 1.0%



各セル: 合格問題数 / カテゴリ問題数、正解率。上下は同じモデル・同じ問題。全9カテゴリを表示。
既存18＋追加middle4 2モデルから、通常1M・1epochの標準/短いBPEを表示除外。元の評価結果は保持。
pass@5 = greedy 1＋sampling 4 (temperature 0.8, top-p 0.95)。［設定復元］は小型loopの既存重みを使用。